Yawar Badri

Senior Systems Engineer & Hardware Architect
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PROFESSIONAL SUMMARY

Full-stack hardware architect specializing in complex systems, hardware pipelines, and emerging energy tech. Proven ability to architect, simulate, and deploy end-to-end solutions spanning from high-power analog frontends and PCB layouts to deeply embedded firmware and deterministic desktop host interfaces.

CORE COMPETENCIES

- **Hardware & EDA:** KiCad, Altium, Cadence, SolidWorks, SolidEdge
- **Simulation & Modeling:** MATLAB, Simulink, LTspice, Qspice
- **Languages & Firmware:** C, C++, C#, Python, STM32, ARM Cortex
- **Test & Measurement:** DSO, MSO, Logic Analyzer, LCR, Megger, PQA, DMM
- **Engineering Methodologies:** FMECA, FMEA, WCCA, RCA, Hardware-in-the-Loop (HIL)

TECHNICAL EXPERIENCE

Pixxel Space | Senior Electrical Engineer *Current*

Leading the end-to-end electronic subsystem design for multiple satellites. Specializing in high-reliability electronics engineered to operate in hazardous orbital environments. Architecting full Electrical Power System (EPS) end to end.

Kashmir University | Technical Advisor

Providing strategic technical architecture guidance to emerging hardware startups, mentoring founders on prototyping-to-production pipelines.

NIT Srinagar | Project Associate (R&D Architecture)

Directed hardware architecture and systems integration for an experimental grid-tied inverter testbed.

SELECTED PROJECTS

Hardware Systems

- **Satellite EPS & EGSE:** Led end-to-end electrical power subsystem design for satellites. Developed critical ground support equipment, logging, and protection instrumentation.
- **10-Channel Power Logger:** Bidirectional DC power logger (100Vmax) featuring custom analog front-end hardware and STM32 USB 2.0 interface. Included hardware design, firmware, calibration, and desktop app.
- **Power Conversion Suite:** Built a 100W linear supply, 100W 2-quadrant supply, and 150W electronic load. Worked on kilowatt-scale grid-integrated PV inverters.
- **DACard-Basic HIL Interface:** Architected a full-stack hardware-in-the-loop testing ecosystem encompassing physical hardware, embedded firmware, host software, and custom Python libraries.
- **Ground Energy Storage BMS:** Designed and deployed comprehensive battery management systems for large-scale energy storage solutions.

Software Tooling

- **CATS (CAN Automated Testing Software):** Used to test hardware with a full command, read, verify, record cycle based on PCAN hardware.
- **KiWay & Transformer Design Tools:** PCB Design routing framework for automating redundant constraints, and internal tools to calculate custom magnetic transformers.
- **sigLib & Gauss-Seidel Load Flow:** C++ DSP library for real-time waveform conditioning, and console-based numerical analyzer for calculating power distribution flows.

EDUCATION

Lovely Professional University (LPU) 2022

Master of Technology (M.Tech) in Electrical Engineering (Power Systems & Renewable Energy) | **Grade: A**

University of Kashmir 2019

Bachelor of Technology (B.Tech) in Electrical Engineering | **Grade: A**

PUBLICATIONS

- *Ultracapacitor Integration with Regenerative Braking Systems in Electric Drive Trains* (ScienceDirect/Elsevier, 2023)

- *Experimental Implementation of Optimally Tuned Multifunctional RLMLS Driven VSc...* (Springer Link, 2023)
- *Application of V2V Energy Sharing in Electric Vehicles with Source Switching* (IEEE Xplore, 2022)
- *A New Modular Multilevel Converter Topology Using Flying Ultra-capacitor...* (Springer, 2021)